Fiscal Year:	FY 2006	**Task Last Updated:**	FY 02/05/2006
PI Name:	Ketterle, Wolfgang Ph.D.		
Project Title:	Towards Precision Experiments with Bose-Einstein Condensates II		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ketterle@mit.edu	**Fax:**	FY 617-253-4876
PI Organization Type:	UNIVERSITY	**Phone:**	617-253-6815
Organization Name:	MIT		
PI Address 1:	77 Massachusetts Ave.		
PI Address 2:			
PI Web Page:	http://cua.mit.edu/		
City:	Cambridge	**State:**	MA
Zip Code:	02139-4307	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/2003	**End Date:**	01/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	0	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JPL
Contact Monitor:	Maleki, Lute	**Contact Phone:**	818-354-3688
Contact Email:	lute.maleki@jpl.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG 3 2937		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock mysteries of the universe obscured by Earth's gravity		
	The picokelvin temperature range will be approached by evaporative cooling followed by adiabatic expansion which lowers the energy even further. Although the theoretical limits of evaporative cooling are in the picokelvin range, the speed of cooling becomes extremely slow because the cooling rate is proportional to the rate of elastic collisions and therefore to the thermal velocity. However, once a Bose condensate has been created at the current critical temperatures of 100 nK to 1 microK, one might adiabatically expand the condensate and thus obtain very dilute condensates with much reduced zero-point energy and interaction energy. The major challenge is that the field curvatures have to be reduced to 10^-5 G/cm^2 while a magnetic field gradient of 8 G/cm is required to compensate for gravity in the laboratory. Such expanded Bose condensates would be coherent matter waves with de Broglie wavelengths approaching 1 mm which are not only very promising for precision metrology, but also for atom optics and conceptional experiments in quantum mechanics.		

Task Description:	A systematic study of the limiting processes in generating such pK samples of atoms will be performed. Possible limitations are the precise control of the magnetic fields and heating due to background gas collisions and vibrations. We expect that these processes will become more critical at lower temperatures. Precise RF (or Raman) spectroscopy can be used to measure fundamental constants like h/m and to test fundamental symmetries like charge neutrality and time reversal. RF spectroscopy of Bose condensates will be performed, both in free fall and confined in a magnetic trap. The first steps will focus on the study of systematic shifts due to the trapping potential and the presence of the Bose condensate at finite density. results from this study will naturally lead to the formulation of a concept how to implement RF spectroscopy with mHz linewidth in the microgravity environment of a space station. The goal is to produce atomic samples with picokelvin temperatures and to improve precision spectroscopy. In microgravity, picokelvin samples would spread out by less than 1 cm/minute and would allow rf and microwave spectroscopy with mHz linewidths, an improvement of three orders of magnitude over earth-based atomic fountain experiments. RF spectroscopy of such ultracold samples will be performed to study shifts and broadening due to the finite atomic density and the presence of the trapping potential. This study should lead to a clear concept of how to perform precision spectroscopy with mHz linewidth on a space station. This holds the promise of greatly improving current atomic clocks and experiments testing fundamental symmetries in physics. This task develops key technologies for preparing picokelvin samples of atoms for precision spectroscopy. At this point, the major limitation for spectroscopy will be gravity. Picokelvin atoms in a microgravity environment holds the promise of greatly improving current atomic clocks and experiments testing fundamental symmetries in physics.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	1. Vortices and Superfluidity in a Strongly Interacting Fermi Gas Quantum-degenerate Fermi gases provide a remarkable opportunity to study strongly interacting fermions. In contrast to other Fermi systems, such as superconductors, neutron stars or the quark-gluon plasma, these gases have low densities and their interactions can be precisely controlled over an enormous range. By varying the pairing strength between two fermions near a Feshbach resonance, one can explore the crossover from a Bose-Einstein condensate (BEC) of molecules to a Bardeen-Cooper-Schrieffer (BCS) superfluid of loosely bound pairs whose size is comparable to, or even larger than, the interparticle spacing. The crossover realizes a novel form of high- superfluidity and it may provide new insight for high- superconductors. Earlier experiments with Fermi gases have revealed condensation of fermion pairs [1-5], but not observed superfluidity. Our observation of vortex lattices directly displays superfluid flow in a strongly interacting, rotating Fermi gas [6]. A strongly interacting cloud of fermions was created by laser cooling, sympathetic cooling with sodium in a magnetic trap, followed by evaporative cooling in an optical trap [3, 5]. The trapped cloud was rotated about its long axis using a blue detuned laser beam (wavelength 532 nm). A two-axis acoustooptic deflector generated a two-beam pattern that was rotated symmetrically around the cloud at a variable angular frequency. Vortex lattices were generated both above and below the Feshbach resonance at 834 Gauss. 2. Fermionic Superfluidity with Imbalanced Spin Populations We have established superfluidity in a two-state mixture of ultracold fermionic atoms with imbalanced spin populations [7]. This study relates to the long-standing debate about the nature of the superfluid state in Fermi systems. Indicators for superfluidity were condensates of fermion pairs, and vortices in rotating clouds. For strong interactions, near a Feshbach resonance, superfluidity was observed for a broad range of population imbalances. We mapped out the superfluid regime as a function of interaction strength and population imbalance, and characterized the quantum phase transition to the normal state, known as the Pauli limit of superfluidity. 3. Sodium Bose-Einstein Condensates in an Optical Lattice Optical lattices have become a powerful tool to enhance the effects of interaction in ultracold atomic systems to create strong correlations and probe many-body physics beyond the mean-field theory [8]. Simply through varying the depth of the lattice potential, one changes the tunneling rate as well as the on-site interaction energy by changing the confinement of the atoms. So far, most optical lattice experiments have been performed with relatively heavy atomic species (e.g. rubidium and potassium) for which the recoil frequencies are lower and lasers are readily available to achieve trap depths of several tens of recoil frequencies at a few tens of milliwatts. For 23Na, high power single-mode lasers are necessary for similar experiments. In this work, we chose to use a dye laser red-detuned by 5 nanometers from the D lines of sodium (589 nm). The spontaneous scattering rate limited the time window of the experiment to less than 50 ms, but was still sufficient to satisfy the adiabaticity condition to explore the quantum phase transition from a superfluid to a Mott insulator [9]. We also observed strong atom losses at various lattice laser detunings, which were interpreted as photoassociation transitions. The particular molecular states responsible for these transitions were identified through theoretical calculations and previous experimental data. 4. Optical Weak Link between Two Spatially Separate Bose-Einstein Condensates The concept of Josephson coupling can be extended to include two spatially separate quantum systems by using intermediate coupling systems. The phase of the coupling may be actively controlled by adjusting the coupling states of the intermediate systems. We have experimentally demonstrate phase-sensitive optical coupling of two spatially separate Bose- Einstein condensates using Bragg scattering [10]. We have studied two condensates in an optical double-well potential, irradiated by two pairs of Bragg beams which couple out beams of atoms propagating to the left or the right, respectively, and these unconfined propagating atoms constitute the intermediate coupling system in our scheme. Depending on the relative phases of the two condensates and the coupling states, we observe only one outcoupled beam propagating to one or the other side, or two identical beams propagating in opposite directions (see figure). This demonstrates phase control of currents and establishes a new scheme to realize Josephson effects with two non-overlapping condensates. 5. Interference of Bose-Einstein condensates split with an atom chip. A major step towards compact matter wave sensors is an atom interferometer on an atom chip. We have used an atom chip to split a single Bose-Einstein condensate of sodium atoms into two spatially separated condensates [11]. Dynamical splitting was achieved by deforming the trap along the tightly confining direction into a purely magnetic double-well potential. We observed the matter wave interference pattern formed upon releasing the condensates from the microtraps. The intrinsic features of the quartic potential at the merge point, such as zero trap frequency and extremely high field-sensitivity, caused random variations of the relative phase between the two split condensates. Moreover, the perturbation from the abrupt change of the trapping potential during the splitting was observed to induce vortices. 6. Observation of Strong Quantum Depletion in a Gaseous Bose-Einstein Condensate Gaseous condensates can be

almost quantitatively described by a single macroscopic wave function shared by all atoms which is the solution of a non-linear Schrodinger equation. The fraction of the many-body wavefunction which cannot be represented by the macroscopic wavefunction is called the quantum depletion. In a homogenous BEC, it consists of admixtures of higher momentum states into the ground state of the system. For typical gas densities, the fraction of the quantum depletion is 0.2% whereas in liquid helium it is approximately 90 %. To bride the gap between Bose condensed gases and liquid helium, we have studied quantum depletion in an optical lattice, which enhanced the atomic interactions and modified the dispersion relation resulting in strong quantum depletion [12]. The depleted fraction was directly observed as a diffuse background in the time-of-flight images. Bogoliubov theory provided a semi-quantitative description for our observations of depleted fractions in excess of 50%.

1. M. Greiner, C.A. Regal, and D.S. Jin, Nature 426, 537 (2003).

2. S. Jochim, M. Bartenstein, A. Altmeyer, G. Hendl, S. Riedl, C. Chin, J. Hecker Denschlag, and R. Grimm, Science 302, 2101 (2003).

3. M.W. Zwierlein, C.A. Stan, C.H. Schunck, S.M.F. Raupach, S. Gupta, Z. Hadzibabic, and W. Ketterle, Phys. Rev. Lett. 91, 250401 (2003).

4. C.A. Regal, M. Greiner, and D.S. Jin, Phys. Rev. Lett. 92, 040403 (2004).

5. M.W. Zwierlein, C.A. Stan, C.H. Schunck, S.M.F. Raupach, A.J. Kerman, and W. Ketterle, Phys. Rev. Lett. 92, 120403 (2004).

6. M.W. Zwierlein, J.R. Abo-Shaeer, A. Schirotzek, C.H. Schunck, and W. Ketterle, Nature 435, 1047 (2005).

7. M.W. Zwierlein, A. Schirotzek, C.H. Schunck, and K. W, Science 311, 492 (2006), published online on Dec. 22, 2005.

8. M. Greiner, O. Mandel, T. Esslinger, T.W. Hänsch, and I. Bloch, Nature 415, 39 (2002).

9. K. Xu, Y. Liu, J.R. Abo-Shaeer, T. Mukaiyama, J.K. Chin, D.E. Miller, W. Ketterle, K.M. Jones, and E. Tiesinga, Phys. Rev. A 72, 043604 (2005).

10. Y. Shin, G.-B. Jo, M. Saba, T.A. Pasquini, W. Ketterle, and D.E. Pritchard, Phys. Rev. Lett. 95, 170402 (2005).

11. Y. Shin, C. Sanner, G.-B. Jo, T.A. Pasquini, M. Saba, W. Ketterle, D.E. Pritchard, M. Vengalattore, and M. Prentiss, Phys. Rev. A 72, 021604(R) (2005).

12. K. Xu, Y. Liu, D.E. Miller, J.K. Chin, W. Setiawan, and W. Ketterle, preprint, cond-mat/0601184.

Bibliography Type:	Description: (Last Updated: 02/05/2006)
Articles in Peer-reviewed Journals	K. Xu, Y. Liu, D.E. Miller, J.K. Chin, W. Setiawan, and W. Ketterle "Observation of Strong Quantum Depletion in a Gaseous Bose-Einstein Condensate." submitted to Phys. Rev. Lett.; preprint, cond-mat/0601184. , Jan-2006
Articles in Peer-reviewed Journals	MW. Zwierlein, A. Schirotzek, C.H. Schunck, and W, Ketterle "Fermionic Superfluidity with Imbalanced Spin Populations." Science 311, 492-496 (2006); published online on Dec. 22, 2005 , Jan-2006
Articles in Peer-reviewed Journals	K. Xu, Y. Liu, J.R. Abo-Shaeer, T. Mukaiyama, J.K. Chin, D.E. Miller, W. Ketterle, K.M. Jones, and E. Tiesinga "Sodium Bose-Einstein Condensates in an Optical Lattice." Phys. Rev. A 72, 043604 (2005) , Jan-2006
Articles in Peer-reviewed Journals	Y. Shin, G.-B. Jo, M. Saba, T.A. Pasquini, W. Ketterle, and D.E. Pritchard "Optical Weak Link between Two Spatially Separate Bose-Einstein Condensates." Phys. Rev. Lett. 95, 170402 (2005). , Jan-2006
Articles in Peer-reviewed Journals	Y. Shin, C. Sanner, G.-B. Jo, T. A. Pasquini, M. Saba, W. Ketterle, D. E. Pritchard, M. Vengalattore, and M. Prentiss "Interference of Bose-Einstein condensates split with an atom chip." Phys. Rev. A 72, 021604(R) (2005). , Jan-2006
Articles in Peer-reviewed Journals	M.W. Zwierlein, J.R. Abo-Shaeer, A. Schirotzek, C.H. Schunck, and W. Ketterle "Vortices and Superfluidity in a Strongly Interacting Fermi Gas." Nature 435, 1047-1051 (2005). , Jan-2006
Articles in Peer-reviewed Journals	C. A. Stan and W. Ketterle "Multiple species atom source for laser-cooling experiments." Rev. Sci. Instrum. 76, 063113 (2005). , Jan-2006